

ASSIGNMENT -1(Subjective) (Structure of An Atom)

1. Calculate the number of neutrons in the product obtained in the following transformation:

$${}_{92}\text{X}^{238} \rightarrow \text{A} + {}_2\text{He}^4$$
2. What is the ratio between the energies of two radiations one with a wavelength of $8000\text{\AA}$ and another with $2000\text{\AA}$.
3. For a broadcasted electromagnetic wave having frequency of 1200 kHz, calculate number of waves that will be formed in 1 km distance (wave number per km).
4. Two bulbs 'A' and 'B' emit red light and yellow light at $8000\text{\AA}$ and $4000\text{\AA}$ respectively. The number of photons emitted by both the bulbs per second is the same. If the red bulb is labelled as 100 watts, find the wattage of the yellow bulb.
5. The Threshold wavelength of a metal is 230 nm. Calculate the KE of the electrons emitted from that metal surface by using UV radiation of wavelength 180 nm.
6. A photon of wavelength $4000\text{\AA}$ Strikes a metal surface, the work function of the metal being 2.13 eV. Calculate the K.E of the emitted electron and velocity of the photoelectron.
7. How many Balmer lines in the spectrum will be observed when electrons return from 7th shell to 2nd shell?
8. The energy corresponding to one of the lines in the Panchen series for H-atom is $18.16 \times 10^{-20}\text{J}$. Find the quantum numbers for the transition which produced this line.
9. $\frac{h}{\pi}$ is the angular momentum of the electron in the $n = \dots\dots\dots$ orbit of He^+ .
10. The electrons identified by n & l ;
 (i) $n = 4, l = 1$
 (ii) $n = 4, l = 0$
 (iii) $n = 3, l = 2$
 (iv) $n = 3, l = 1$
 can be placed in order of increasing energy from lowest to highest?

11. The speed of this dust particle (mass = 10^{-3} g) is measured with the uncertainty of $\frac{3.313}{\pi} \times 10^{-3}$ m/s. The minimum uncertainty in position of the dust particle (in order of 10^{-26} m) is :
12. The circumference of the second orbit of an atom or ion having single electron, is 4 nm. The de Broglie wavelength of electron (in nm) revolving in this orbit is:
13. The uncertainty in position and velocity of a particle are 10^{-10} m and 5.27×10^{-24} ms⁻¹ respectively. Calculate the mass of the particle
($h = 6.625 \times 10^{-34}$ Joule Sec.)
14. If the energy released when an electron jumped from the 4th orbit to the 3rd orbit of hydrogen is 'x' then what would be the energy difference when it jumps from the 3rd orbit to the 2nd orbit?
15. A particular atom has the 4th shell as its valence shell. If the difference between the number of electrons between K and N shells and L and M shell is zero, find the atomic number of the element and electronic configuration of its stable ion.
16. A stable unipositive ion of an element contains three fully filled orbits. What is the atomic number of the element.
17. The average atomic mass of two isotopes with mass number A and A + 2 is A + 0.25. Calculate the percentage abundance of the isotopes.
18. The radius of nth orbit of single electron species is $0.132 n^2 A^\circ$. Identify the element.
19. What is the frequency of light emitted when an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit?
20. An electron having an angular momentum of 1.05×10^{-34} joules jumps to another orbit such that it has an angular momentum of 4.20×10^{-34} joules. Explain the possible transitions.
21. If ${}_x^yA^{+1}$ or ${}_{x-1}^{y-2}B^{+1}$ were to be used instead of α -particles in Rutherford's experiment, which would be better and why?